

Practical: 01

Aim: Design a Logo for an E-Commerce app.

What is a logo?

Amazon's orange arrow stretches from A to Z, representing the extensive range of their products and also the dimpled smile of their happy customers.



In a nutshell, a logo is a mark made up of text and images that is adopted by an organization to identify its offering. Ideally, a logo will present a brand to an audience while also differentiating it from the competition. It will be packed with meaning and perfectly communicate the industry, services, demographic and values of a company so that consumers can speedily decide if it's for them.

What makes a good logo?

However you might feel about McDonald's, its golden arches are the stuff of logo legend. A good logo tells the story of your brand using design elements like color, fonts, imagery and shapes. It will be easy to distinguish against competitors and have an impact wherever it's seen. A good logo is also translatable across every medium, and is clearly distinguishable whether it's on a business card, or an Instagram logo.

“It can be as pretty as you like, but if it doesn't mean anything and doesn't communicate anything, then it's not going to be a good logo. It needs to be effective at communicating the brand—its purpose and personality—and it needs to do that in a very simple, clear way. It also needs to be memorable and recognizable.”

The design elements included in a logo

FedEx makes clever use of negative space between its 'E' and 'x'.



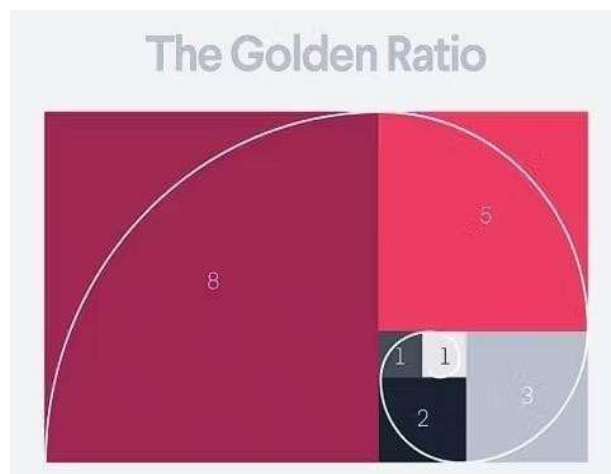
The basic ingredients that make up a logo are:

- Colors
- Typeface/font
- Line
- Shape
- Negative space
- Graphics/icon
- Layout

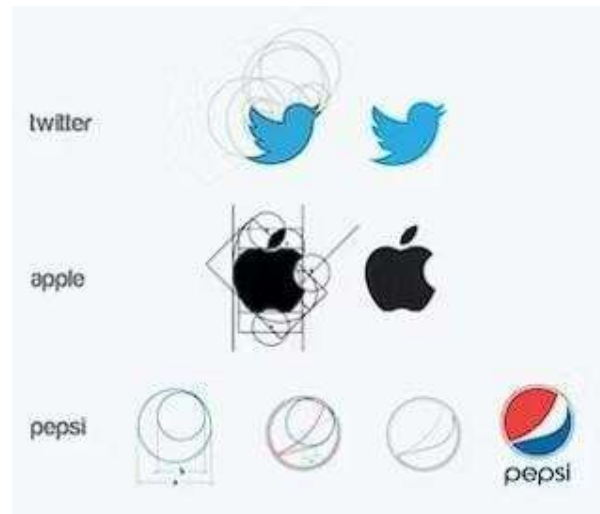
What is the Golden Ratio?

The Golden Ratio is the number used when two quantities are divided in a way that their ratio is the same as the ratio of their sum to the larger one of the two quantities. That number is 1.618, also called Phi.

The easiest way to demonstrate this is by using the Fibonacci Sequence. Without going into too much detail, this sequence is the sum of the two numbers before it. So: 0,1,1,2,3,5,8,13,21...forever and ever (and ever). Back in the day, the Greeks used the Fibonacci Sequence to form a visual pattern to aid their designs. When you turn the sequence into squares and lay them side-by-side to create rectangles, a spiral (called the Golden Spiral) starts to form.



When designing a logo, you can even imagine the Fibonacci sequence as a series of circles, then rearrange them to form a grid as the foundation for your logo design. This is the basis of many logos, including the Twitter bird.



Logo:



Practical: 02

AIM: Design an email that showcases a promotional offer of the e-commerce app.

Promotional emails have become a powerful tool for eCommerce businesses to communicate their offerings, drive sales, and build customer relationships.

What Are Promotional Emails?

Promotional emails are a form of targeted marketing communication delivered to a company's subscribers or customers via email. These emails aim to promote products, services, events, or special offers to encourage recipients to take a desired action.

Through persuasive content, visually appealing designs, and strategic calls to action, promotional emails are a direct channel for businesses to engage their audience, boost brand awareness, and drive conversions.

Why Are Promotional Emails Important?

There are several reasons why you should consider using promotional emails.

1. Build Trust & Get Customers Engaged

Promo emails build trust with your customers by providing them with real value, whether that be a discount, free information, or otherwise. This makes promotional emails a great tool for converting browsers into customers who will trust you and purchase your product or service.

Promotional emails also build engagement by giving people an incentive to engage with your content.

In fact, promotional emails that contain both content and incentives (like lead magnets) see the highest click-through rates (CTR), conversions, and customer loyalty.

2. Save Money

Promo emails are cost-effective promotional tools because they target audiences that have already shown interest in your products or services through their browsing behaviours. Plus, most email service providers have reasonable fees that are more affordable than other marketing strategies, such as pay-per-click advertising.

3. Boost Your ROI

Since promotional emails are targeted toward audiences that have already shown interest in your business or products, promotional emails have the highest return on investment (ROI) of all promotional tools. This is because promotional emails provide real value to the recipient, giving them a greater sense of trust and appreciation for your business.

Common Qualities of Promotional Emails

1. Promotional Emails Establish Positive Relationships:



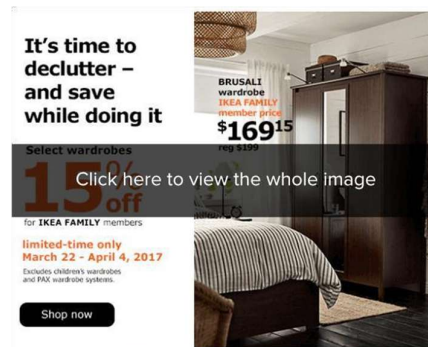
2. Promotional Emails Offer Something of Value:



3. Emails That Know Its Audience:



4. Emails That Are Easy to Read



Promotional Email:



WINTER SEASON!

* Winter Sale: Shop Crazy with Hot Deals! *

The winter sale has arrived, and so have the hottest deals of the season! *

**SHOP FAST.
GET FAST!**

Shop now

NEW DEALS



Groceries



Gifts



Clothing



More

BACK IN OFFER

Don't miss out on these fantastic seasonal offers! Visit our store now and discover discounts on select items: &

fastcartsupport!7@gmail.com

Practical: 03

AIM: Design brochure that showcases different features of the e-commerce app.

When it comes to brochures, it's all about the design. A great design will compel your audience to read all about what you're doing. A less-than-stellar design will end up in the trash can.

So how, exactly to design a brochure?

Before you start designing

- Know your brand personality
- Define your ideal customer
- Develop your message
- Determine your metrics for success
- Set your budget

Designing your brochure

- Remember your brand design standards
- Design with the reader in mind
- Choose your brochure type
- Gather your copy and images
- Find your style
- Pick the perfect CTA

Evaluating and printing

- Evaluate your design
- Choose a printer
- Explore print options

Before you start designing your brochure

The key to creating an ah-mazing brochure actually starts before you design by knowing your brand personality, message, and target customer—you'll make the design process go a lot more smoothly.

1. Know your brand personality:

Knowing your brand personality is a must. If you don't know your brand inside and out, all of your branding materials—including your brochures—will feel disjointed and unclear

2. Define your ideal customer

Before you start designing your brochure, get crystal clear on who you're designing for. Different audiences require different designs, and if you're not clear on your audience, you run the risk of making the wrong design choices.

Ask yourself:

- Who is my ideal customer?
- What kind of information are they looking for?
- Are they more likely to respond to more images or more text?
- What kind of copy do they expect? (i.e. Corporate or conversational? Humorous or serious?)
- What can I do to best grab their attention?

When you know who you're designing for, use that to steer your design decisions. You'll end up with a brochure that feels true to them, which will help up your chances of success.

3. Develop your message:

It's so, so important that you define your message. You need to know what you're going to say in your brochure and how you're going to say it before you even think about getting a design in place. Because your message is the most important thing. It all comes back to knowing your customer.

If you don't have a strong, clear message that speaks to them in language and images they can relate to, it doesn't matter what design you come up with.

4. Determine your metrics for success:

determine your metrics by defining what you're hoping to get out of your brochure.

- Are you looking to drive people into a retail location? Include a coupon or voucher and measure how many of them are redeemed at your store.
- Are you trying to drive people to your website? Include a custom URL on the pamphlet and track the number of visitors during the campaign.
- Are you trying to build buzz around a new product launch? Include a CTA to sign up for your email list to get updates and see how large your list grows during the campaign.

5. Set your budget:

Come up with a budget-per-print and start making some decisions based on what's most important. Knowing how much cash you have on hand for the design and printing process will help you make the best decisions for your budget and squeeze the most out of every dollar.

Brochure:



"Your Fast Track to Online Shopping"

Where to visit us
LINK
www.fastcarttsite.com
ADDRESS
Mbit, V V nagar City, 388080
CONTACT US
1234567890
hello@fastcartsite.com
www.fastcarttsite.com
CONNECT WITH US
[@fastcartsite](https://www.instagram.com/fastcartsite)

FASTCART
Cart It, Get It Fast!
www.fastcarttsite.com

Get 15% OFF your first order with code fastcart!

Practical: 04

AIM: Create User Personas, User Stories, and Use Case Diagrams to define a Problem Statement, Scope, and understand the problem.

Designers and engineers often face challenges using CAD software due to complexity, steep learning curves, and limited collaboration features. Additionally, managing versions and sharing designs with teams can be inefficient, leading to delays and errors in product development.

Solution:

Develop a modern CAD platform that offers an intuitive interface, robust design tools, cloud-based collaboration, version control, and seamless integration with other engineering tools.

Scope:

- 2D and 3D modeling tools
- Version control and design history
- Real-time multi-user collaboration
- Import/export support for popular CAD formats
- Cloud storage and backup
- Integration with simulation and analysis tools
- User management and role-based access
- Tutorials and help system

User Personas

1. Alex Johnson – Mechanical Engineer

Age: 30

Occupation: Mechanical Design Engineer

Tech Proficiency: High

Pain Points: Complex UI in traditional CAD tools slows down work; hard to share designs easily with the team.

Goals: Wants a powerful yet intuitive tool with collaboration to accelerate design cycles.

2. Priya Singh – CAD Drafter

Age: 24

Occupation: Junior CAD Technician

Tech Proficiency: Moderate

Pain Points: Struggles with steep learning curve and confusing software interfaces.

Goals: Needs an easy-to-use platform with helpful tutorials and guidance.

3. Michael Lee – Project Manager

Age: 40

Occupation: Engineering Project Manager

Tech Proficiency: Moderate

Pain Points: Difficulty tracking design changes and version control across teams.

Goals: Wants transparent collaboration, version control, and ability to monitor progress.

User Stories

- As an engineer, I want to create and edit 3D models efficiently, so that I can develop accurate designs quickly.
- As a drafter, I want guided tutorials and templates, so that I can learn the software fast and produce quality drafts.
- As a project manager, I want to track revisions and changes in real-time, so that I can monitor progress and ensure quality control.
- As a user, I want to collaborate with team members on the same design simultaneously, so that we can reduce communication delays.
- As an engineer, I want to export designs to various file formats, so that I can share with different manufacturing tools.

Use Case Diagram

Actors:

- Engineer
- Drafter
- Project Manager

Admin:

Use Cases:

- Engineer
 - Login/Register
 - Create/Edit 2D and 3D Models
 - Save & Version Control
 - Collaborate with Team
 - Export Designs
- Drafter
 - Login/Register
 - Use Templates
 - Follow Tutorials
 - Create Draft Designs
 - Submit for Review
- Project Manager
 - View Project Status
 - Track Design Versions
 - Approve/Request Changes
 - Manage User Access

- Admin
- Manage Users & Roles
- Manage Licenses
- Oversee System Health

Practical: 05

Aim: User research: Document & Conduct user, market, product, and competitive research to find possible solutions.

Certainly, let's break down user research into different types and provide examples for each:

1. User Research

Objective: Understand the needs, pain points, workflows, and preferences of potential CAD users.

Methods:

Interviews: Conduct in-depth interviews with engineers, drafters, project managers.

Surveys: Use structured questionnaires to gather quantitative data on usage habits and feature priorities.

Contextual Inquiry: Observe users in their actual work environment using current CAD tools.

User Personas: Create detailed personas (as previously done) based on research findings.

Key Questions:

What are the biggest challenges when using current CAD tools?

What features do users rely on most?

How do they collaborate on designs?

What devices/platforms do they prefer?

What frustrates them in their workflow?

2. Market Research

Objective: Identify the size, trends, and demands in the CAD software market.

Methods:

Industry reports: Review reports from Gartner, Forrester, IDC, or niche engineering software analysts.

Market surveys: Analyze data on CAD software adoption by industries (automotive, aerospace,

architecture).

Trend analysis: Look at emerging tech trends like cloud CAD, AI-assisted design, VR/AR integration.

Insights to gather:

Market size & growth rate for CAD software.

Customer segments most actively adopting CAD solutions.

Pricing and business models (subscriptions, licenses).

Regulatory or compliance requirements impacting CAD tools.

3. Product Research

Objective: Understand the features, strengths, and weaknesses of existing CAD products.

Methods:

Feature analysis: List key features of top CAD products (e.g., AutoCAD, SolidWorks, Fusion 360).

User reviews: Analyze reviews on platforms like G2, Capterra for user satisfaction and complaints.

Hands-on testing: Trial popular CAD tools to evaluate usability, collaboration, performance.

Key Focus Areas:

- Usability and learning curve.
- Collaboration features.
- Integration capabilities.
- Platform compatibility (Windows, Mac, Web).
- Pricing and licensing flexibility.

4. Competitive Research

Objective: Identify direct and indirect competitors, understand their positioning, and find gaps or opportunities.

Methods:

Competitor profiling: Create profiles of top competitors with their USPs, weaknesses.

SWOT analysis: Strengths, Weaknesses, Opportunities, Threats for each competitor.

Market positioning: Understand competitor target users, pricing, marketing strategies.

Questions to answer:

Who are the top 3-5 competitors in the CAD space?

What features or services do they excel in?

What are their common user complaints?

Are there underserved niches or features not covered well?

Practical: 06

AIM: Ideation: Create sketches and low-fidelity wireframes of the scoped solution.

What is low fidelity wireframing?

Low-fidelity wireframing is a preliminary stage in the design process for digital products like websites and apps. It involves creating a basic, simplified visual representation of the product's layout and functionality.

These wireframes are often rough sketches that lack details such as color, images, or interactive elements. They serve as a means to outline the product's user flow and basic design concepts, helping both the design team and stakeholders understand the core functionality and user experience. Low-fidelity wireframes are valuable for gathering feedback early in the design process and refining the product's concept before moving on to more detailed design work.

What details should be in a low-fidelity wireframe? Elements can vary between wireframes, but below are the most common elements lo-fi wireframes have.

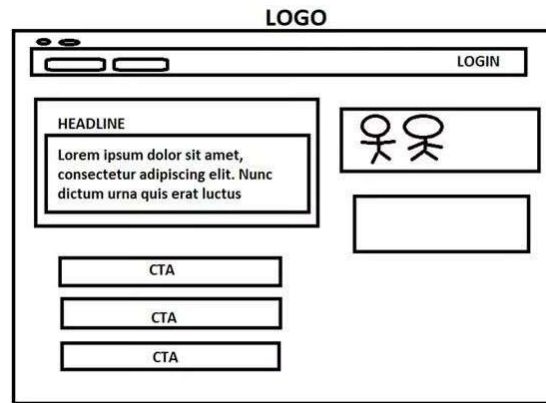
- **Company logo:** Adding your logo signifies your ownership of the product design.
- **Search field:** A search bar or field will simplify navigating your website or mobile app.
- **Breadcrumbs:** Since low-fidelity wireframes do not have clickable elements, you will use breadcrumbs to show the connections between elements. For instance, a breadcrumb will show the page a user will see after clicking a particular element.
- **Headers:** Headers give structure to the content of a wireframe by labelling each element.
- **Placeholders:** You can use shapes to represent buttons and text boxes. You can also use image and text placeholders to represent content on a page.

4 steps for low-fidelity wireframing:

Whether doing low-fidelity wireframing with pen and paper or a digital tool, follow these steps to get the best results:

- **Define your product's scope and purpose:**
Before creating a low-fidelity wireframe, it's essential to establish a clear understanding of your product's scope and purpose. Defining these aspects will enable you to create wireframes that accurately represent the intended functionality and user experience of your product. Additionally, this clarity will make it easier to identify and incorporate the necessary features and functionality in your low-fidelity wireframes to align with your product's purpose.
- **Create a basic layout:**
Create a basic layout of your digital product using simple shapes and placeholders to represent different elements such as content and tabs. Then, create a storyboard that

illustrates the low-fidelity wireframe and mockup in a step-by-step flow, showcasing the user's journey through your website or app.



The storyboard should consist of simple sketches outlining the user's interactions and steps to achieve their objectives within your application.

- **Prioritize usability:**

Your low-fidelity wireframe should be easy to understand at a glance. You can achieve this by prioritizing usability and functionality. Without verbal explanation, stakeholders should be able to see and understand the outcomes of interacting with any elements in the wireframe.

- **Iterate and Refine:**

Share low-fidelity wireframes with team, users, and stakeholders for feedback. Use feedback to improve wireframe iteratively. User input is vital for identifying design shortcomings. Once low-fidelity wireframe is refined, create a high-fidelity version to visualize design elements and assess its effectiveness in meeting its purpose.

Low-fidelity wireframe:

The wireframes are organized into two columns and three rows.

- Top Row:**
 - Login:** Includes fields for Email, Password, and a "Remember me" checkbox. Buttons for "Sign in", "Don't have an account? Create Account", "or continue with", "Google", "Microsoft", and "Try Demo Version".
 - Create Account:** Includes fields for Full Name, Email, Company (optional), Password, and Confirm Password. A checkbox for "I agree to the terms of services & privacy policy". Buttons for "Create Account" and "Already have an account? Sign In".
 - Projects:** Includes a "Search Project" field, "Recent Project" list, "Create New Project" button, and a grid of project thumbnails labeled Project 1 through Project 8.
- Middle Row:**
 - Create Page:** Includes a "Project Name" field, "Select Draw Move Exclude Measure" options, a large drawing area with "2D/3D Canvas" and "Touch to pan, Pinch to zoom" instructions, and a "Zoom 100%" indicator.
 - Collaboration:** Includes a "Workspace Collaboration" header, "chat" and "Collaborators" sections, a list of collaborators (Vish, Kartik), a "Type a msg" field, and a "Send" button.
 - Collaboration (Right):** Includes a "Workspace Collaboration" header, "chat" and "Collaborators" sections, a "Currently online" list with status indicators (Vish: Editing layer 2, Kartik: Viewing), a "Recently Active" list, and a "+invite collaborators" button.
- Bottom Row:**
 - Setting & Profile:** Includes a "Dashboard Profile & Settings" header, a user profile card (Vish, nishagmail.com, Professional Plan, Engineering Corp - Senior CAD Designer), "Settings" (Theme: Light/Dark, Default Units: metric/imperial, Auto save: Every 5 min), "Storage" (45.2 GB of 100 GB used), and a "Manage storage" button.
 - Settings & Profile (Right):** Includes a "Notifications" section (Email notification, Collaboration alert, Version update, Marketing emails), "Change password", "Manage Billing", "Download my Data", "Help & support", and "Account Actions" (Deactivate Account, Delete Account).

Practical: 07

AIM: Execution: Create High-Fidelity Mockups & Prototypes from the wireframes.

What is high-fidelity prototyping?

High-fidelity or hi-fi prototyping helps verify a digital product's design and functionality are ready for the market. It involves building high-fidelity prototypes - preliminary, interactive prototypes that closely simulate a final product's look and functionality.

Hi-fi prototypes for digital products are typically computer-based mockups that support user interaction. The interactive prototype will have realistic graphics, animations, clickable buttons, menus, and other elements that will appear in the finished product.

High-fidelity prototyping enables stakeholders and users to test the prototype for value and detect UX/UI problems early. Without it, collecting accurate human performance data on design and functionality is challenging, leading to a risky blind product development process. Neglecting this can result in a final product with usability and aesthetic issues, potentially causing an 88% user retention drop for websites and apps.

Why is high-fidelity prototyping important?

High-fidelity prototypes accurately represent how your final product's user interface and user experience will look. Since the prototype looks and behaves like live software, users can interact with it normally and provide clear feedback about their experience. Their feedback will provide valuable insights into product design elements you should remove, keep, modify, or fix.

You can perform high-fidelity prototyping at various stages of product development. However, since it's expensive and time-consuming, most design teams save hi-fi prototyping for the final stages of user testing before launch. If you can afford it, you can make hi-fi prototypes during the early stages of product development to test design concepts and user interface variations.

What if I choose to skip high-fidelity prototyping?

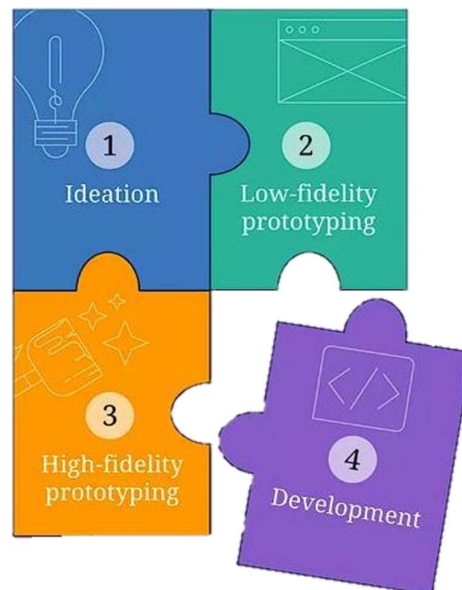
If you do, you will save money during product development and feel like the smartest person in the room. However, down the road, you are likely to face the following problems, which will cost far more than you saved:

1. **Design flaws and usability issues:** Skipping high-fidelity prototyping can lead to undetected design flaws and usability issues. Users will experience these issues after you release your finished product, resulting in low user satisfaction and a tarnished brand image. Not many brands, especially new ones, can recover from a bad product launch.
2. **Lost revenue:** Instead of making money from your launched product, you could end up spending money fixing problems you could have spotted during high-fidelity prototyping. Investing in high-fidelity prototyping will reduce the risk of facing such a costly problem.
3. **Lack of user interest:** High-fidelity prototyping will reveal if your product is something your target users want. It will help you avoid investing time and money to build a product you cannot sell.

In summary, skipping high-fidelity prototyping increases your likelihood of building a less effective, sellable, and user-friendly product. SaaS brands that make this mistake end up losing revenue and customer trust.

How does high-fidelity prototyping fit into the development process?

Product development processes typically vary between SaaS businesses and products. However, the process will still follow a general path, starting at ideation and ending with a product launch. Let's look at the stages of SaaS product development, so you'll see where and how high-fidelity prototyping fits into the process.



1. Ideation:

Ideation is production team members brainstorming what the product should be and how it should work and look. Besides brainstorming ideas with your team, you can get more ideas through target market research, such as customer interviews.

Your research will provide insights into the problems your target audience wants solved and how they want the solution to work. You can use the insights to come up with a product design that features the functionalities your target users desire.

2. Low-fidelity prototyping:

After identifying promising ideas and eliminating unnecessary ones, you can start developing concepts for the product. Developing product concepts involves creating low-fidelity prototypes, such as simple wireframes and mockups that represent how your product will look.

Your wireframes and mockups can be as simple as a rough sketch paper prototype.



Alternatively, build your rough prototype with a low-fidelity prototyping tool like Balsamiq.

Low-fidelity prototyping provides a picture of how your proposed product will look so you can decide if your design ideas make sense. You can also use the prototype to test and refine basic design concepts. Once your low-fidelity prototype makes sense, you can transition to high-fidelity prototyping.

3. High-fidelity prototyping:

Low-fidelity prototyping helps you avoid investing time, money, and resources into building a flawed high-fidelity prototype. Unlike low-fidelity prototypes, which are not interactive, high-fidelity prototypes are interactive, meaning you can click buttons and navigate the interface.

Since hi-fi prototyping requires more money and time, do not start it until lo-fi prototyping has given you a clear picture of how your UI/UX design should look. UI/UX design will include the visual design of your product and the layout and function of your product's interactive elements. Once you have these details, you can build a high-fidelity prototype to test:

- UI elements, color schemes, and copy
- Transitions, animations, and effects

Share the prototype with users and stakeholders to see how users respond to the design and confirm if the product delivers value. After gathering user feedback with your first hi-fi prototype, you can make other iterations that incorporate the feedback.

4. Development:

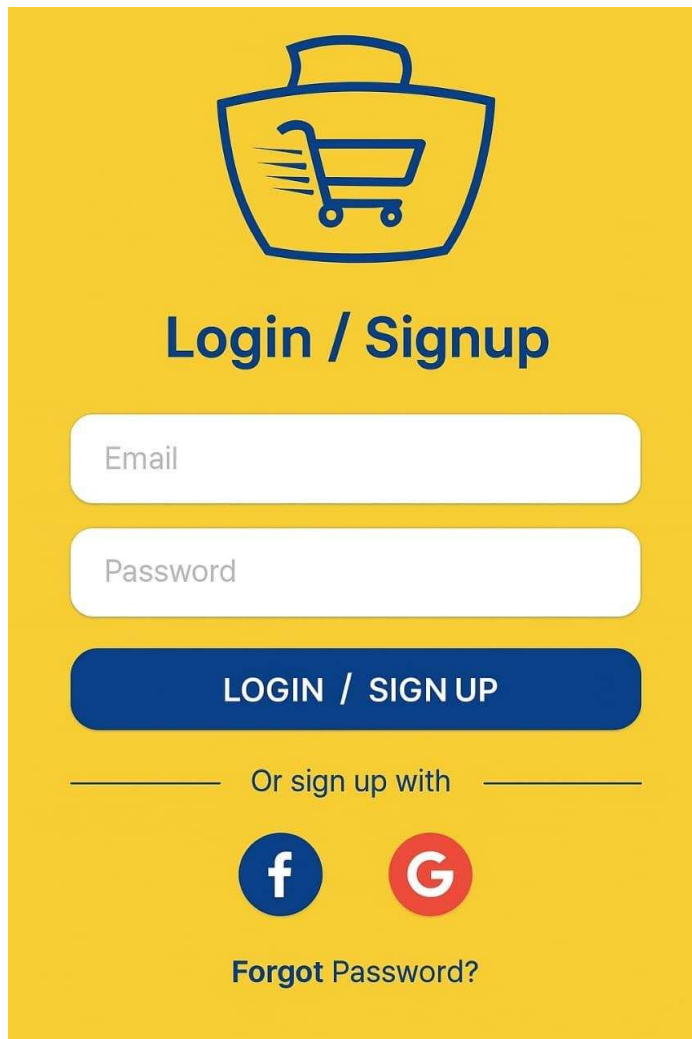
Once you have a high-fidelity prototype that effectively satisfies user needs and meets design requirements, you can start developing version one of your product. The development process will involve building and optimizing your product for the market.

After fully developing and testing the product, you can launch and monitor it. Monitoring the product's performance in the hands of end users will help you identify opportunities for improvement and additional features.

High-fidelity prototyping:

If it's your first time hearing about high-fidelity prototypes, you probably would like to know what they look like. Below are four screenshots from a high-fidelity prototype we created for our Hazel Boutique mobile app.

High – Fidelity page 1:

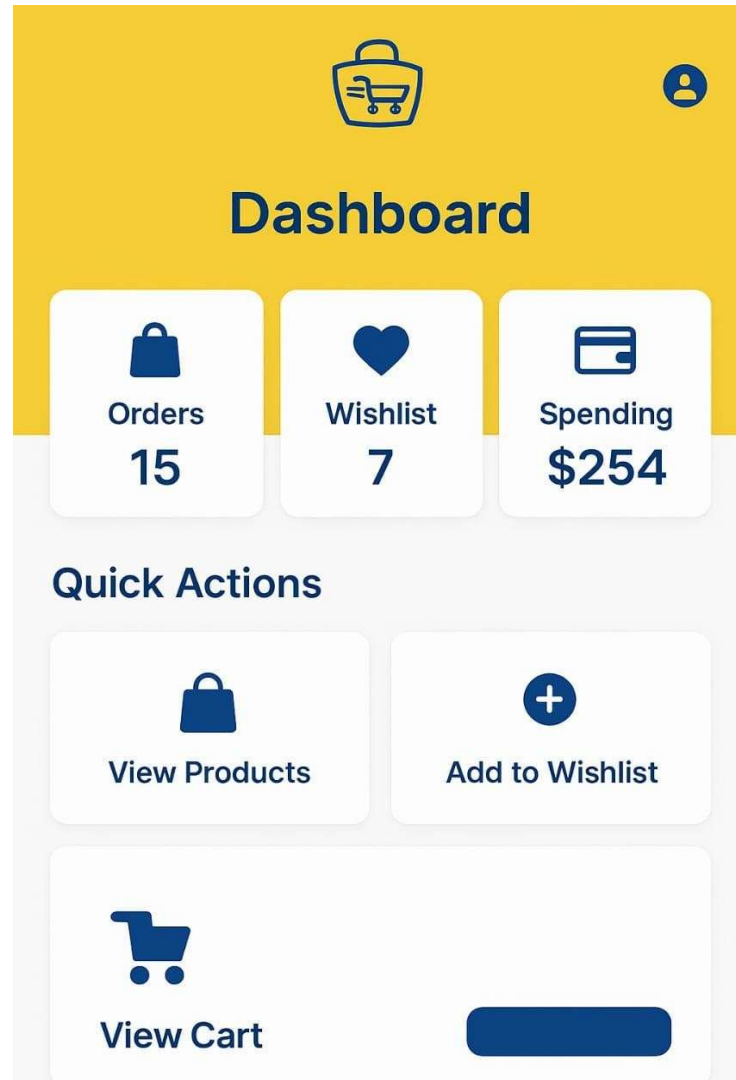


The screenshot shows a login/signup screen with a yellow background. At the top is a blue icon of a shopping basket with a shopping cart inside. Below the icon is the text "Login / Signup" in bold blue font. There are two white input fields with rounded corners: the first is labeled "Email" and the second is labeled "Password". Below these fields is a blue button with white text that says "LOGIN / SIGN UP". Under the button is a horizontal line with the text "Or sign up with" in the center. Below this line are two circular icons: a blue one with a white "f" for Facebook and a red one with a white "G" for Google. At the bottom is the text "Forgot Password?" in bold blue font.

The above screenshot shows the welcome page of the high-fidelity prototype. It resembles a finished product and is more detailed than the below low-fidelity prototype we made for the same product.

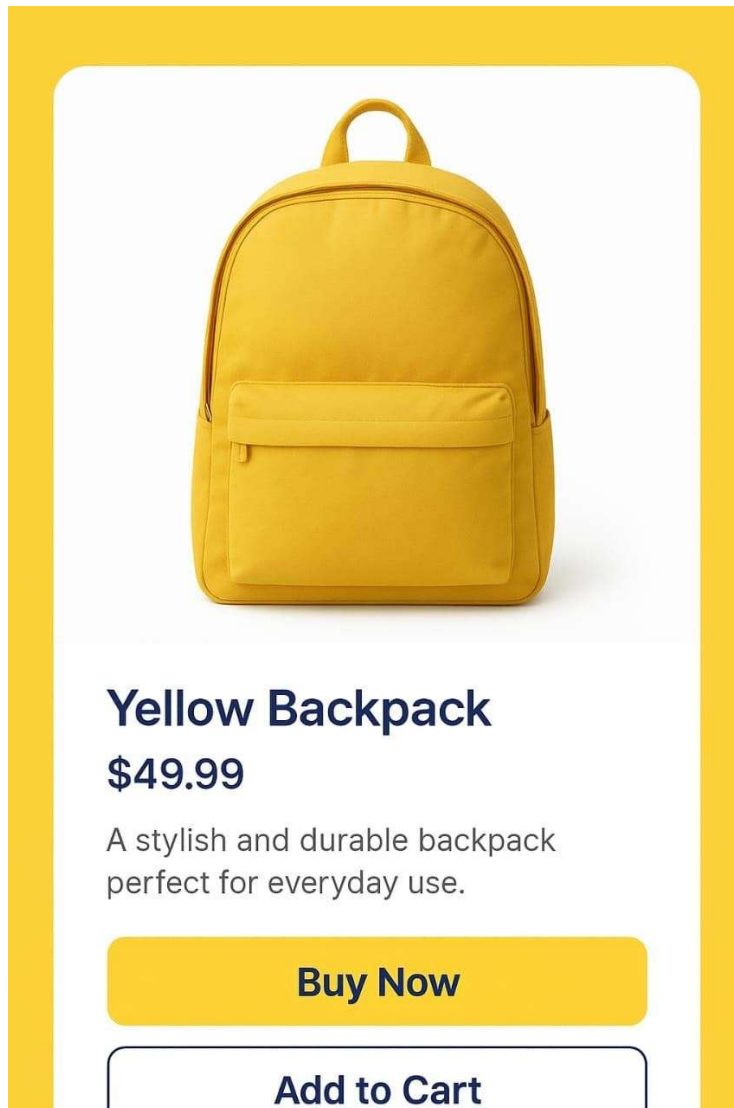
The elements seen in the hi-fi prototype will all appear in the finished product. Unlike in a lo-fi fidelity prototype, these elements aren't just for show. You can click on most of them to experience user flow within the app.

High – Fidelity page 2:



The page of the high-fidelity prototype clearly shows how users can browse categories for various clothes collection using smart device.

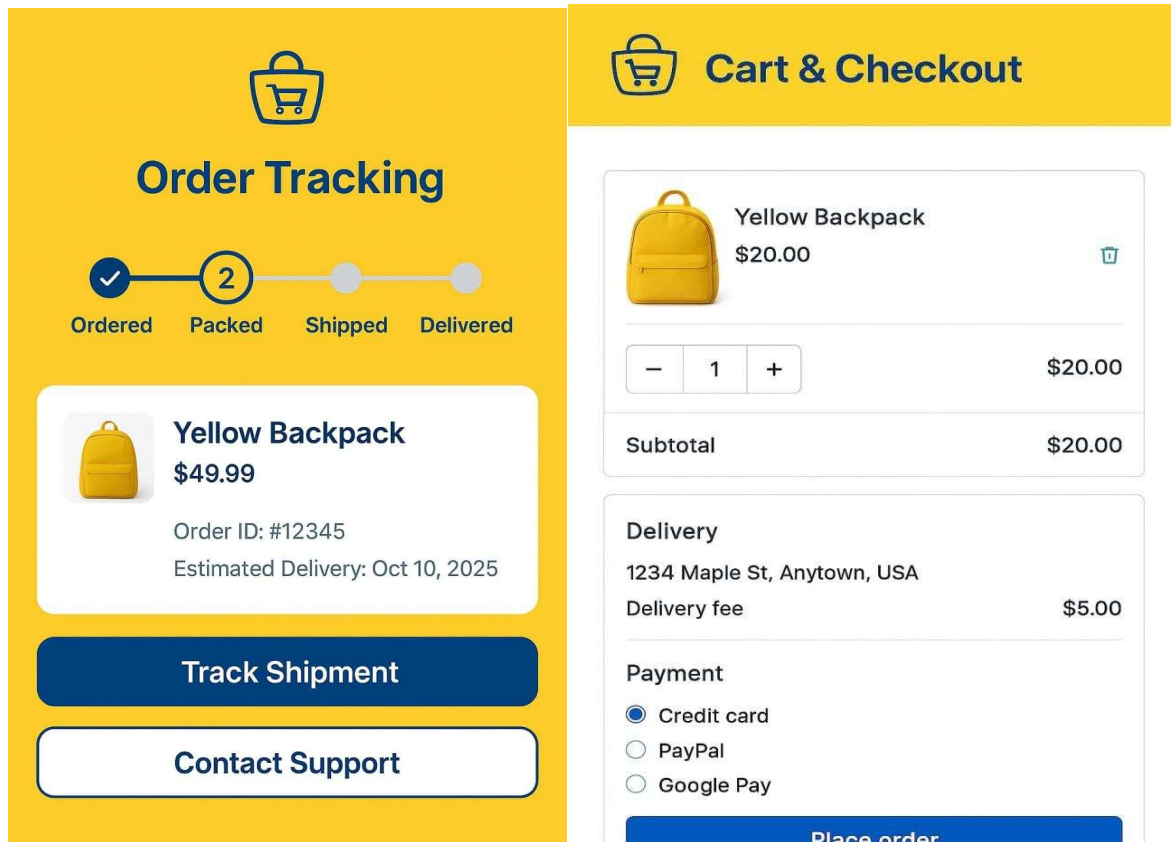
High – Fidelity page 3:



This example shows what happens after clicking the profile button. On this page, you can see your details.

High – fidelity page 4:

The final example shows the last page of the prototype app, which shows the products in the cart.



Practical: 08

AIM: Figma basics: Creating basic responsive elements like buttons, input elements, etc. to understand frames, groups, layout, constraints, texts, vector, color palette, etc.

Creating basic responsive elements

Create a New Figma Document:

Open Figma and create a new document (File > New File) or open an existing one.

Create a Frame:

Click on the Frame tool in the toolbar (or press the 'F' key) and draw a frame on the canvas. Frames are containers for your design elements.

Button Element:

To create a basic button, use the Rectangle tool (R) to draw a rectangle inside the frame. This rectangle represents the button's background.

Style the Button:

Select the rectangle and adjust its properties in the right sidebar:

- Fill: Set the fill color for the button.
- Corner Radius: Round the corners to make it look like a button.
- Size: Set the width and height of the button.

Text Element:

Use the Text tool (T) to add text to the button. Click on the canvas, and a text box will appear. Type in the button label (e.g., "Submit" or "Click Me").

Customize the text properties in the right sidebar:

- Font: Choose a suitable font.
- Size: Set the text size.
- Color: Define the text color.

Grouping:

Select both the rectangle and the text (click and drag to select both).

Right-click and choose "Group" or use the shortcut (Cmd/Ctrl + G) to group them together.

This keeps the button and text aligned.

Responsive Design:

To make the button responsive, you can use constraints:

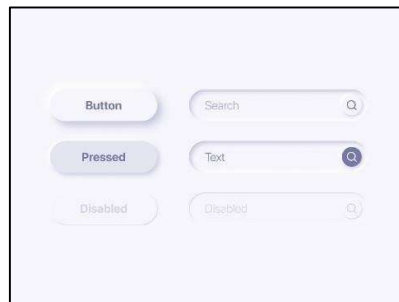
- Select the button group.
- In the right sidebar, click the "Constraints" icon (looks like a chain link) and set constraints to determine how the button resizes within its frame.

Save and Export:

Save your Figma document.

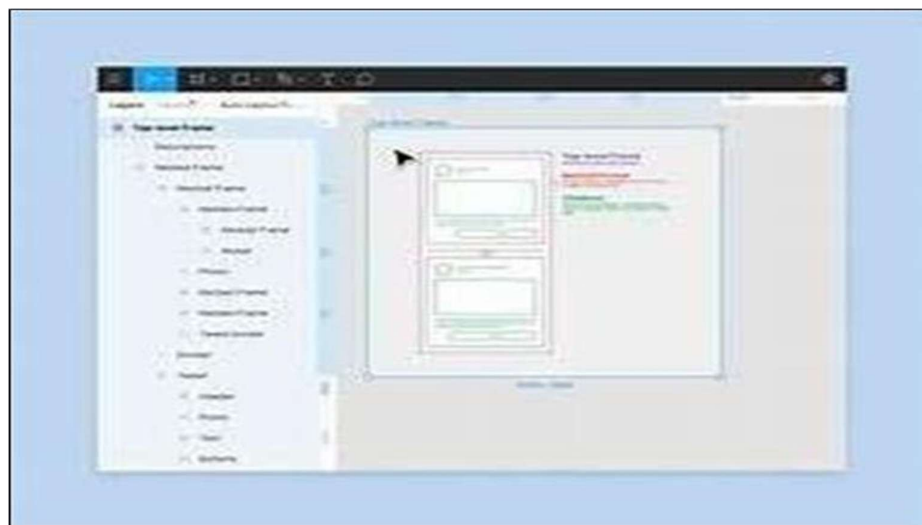
To export your design, go to File > Export and choose your preferred format (e.g., PNG, SVG) and settings.

Example:-



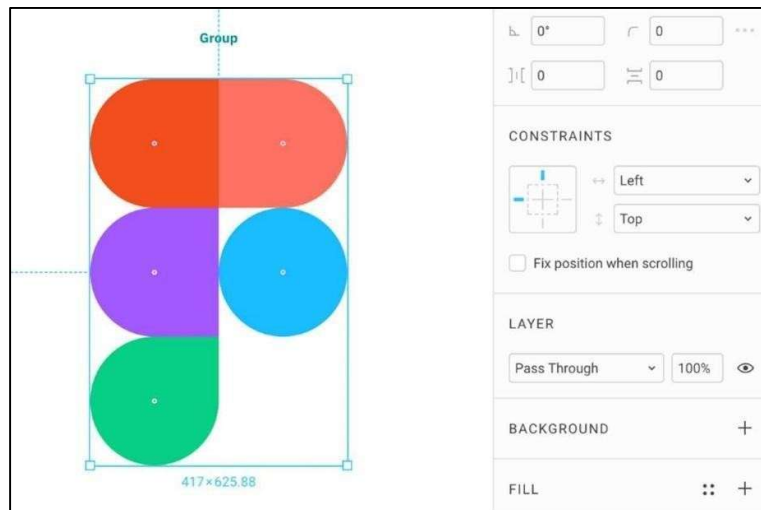
Frames:

- A frame is a container that holds design elements like buttons, input fields, and text.
- Frames help organize and structure your design. You can create frames by drawing rectangles or selecting objects and using the "Frame" option in the toolbar.
- Frames can be resized and used to define the boundaries of your responsive elements



Groups:

- Groups are used to combine multiple objects into a single entity. For example, you can group a button and its label text to keep them together.
- Grouping makes it easier to manipulate and move multiple elements at once.



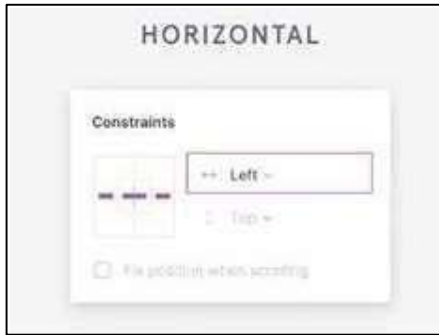
Layout:

- Figma provides layout grids and auto-layout features to help create responsive designs.
- Layout grids allow you to create columns and rows to align and distribute elements evenly.
- Auto-layout is a feature that helps elements resize and reposition automatically when the container (frame) changes size. This is especially useful for responsive design.



Constraints:

- Constraints control how elements respond when the parent frame changes size.
- You can set constraints for each element within a frame to specify how they should resize or stay anchored to certain positions when the frame resizes.



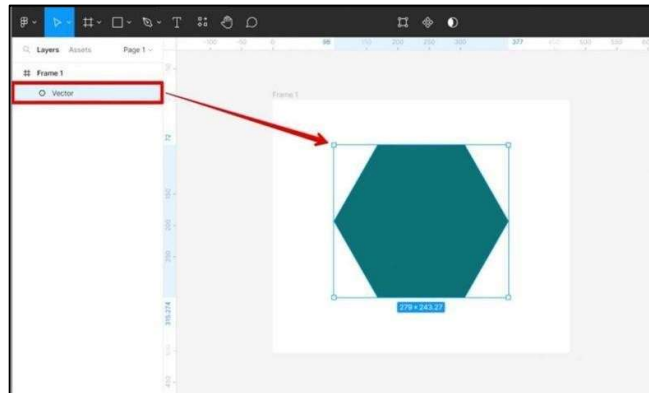
Text:

- Figma allows you to add text layers for headings, labels, or any other text content.
- You can format text with various fonts, sizes, styles, and alignment options.
- Text layers can be edited directly within the canvas or in the properties panel.



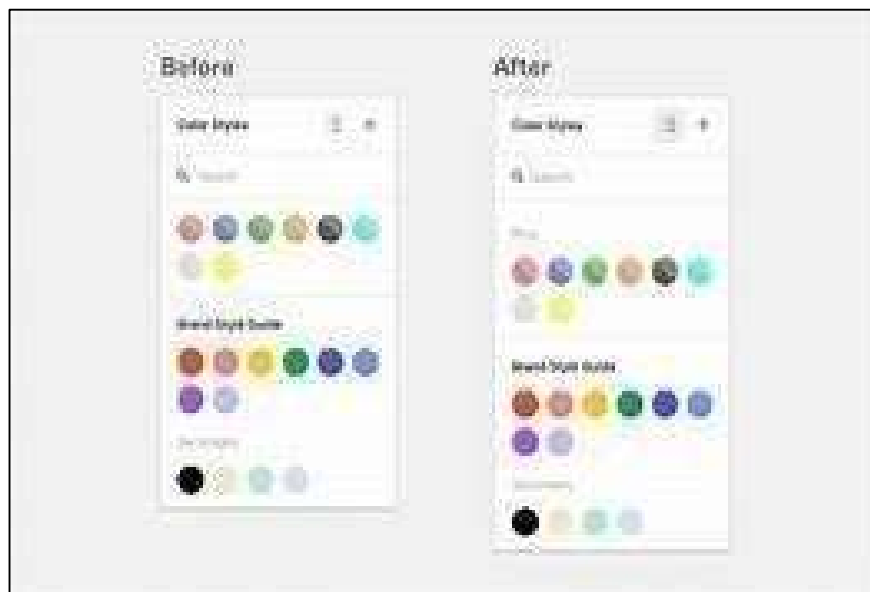
Vector:

- Figma supports vector graphics, which are scalable and retain quality at any size.
- You can draw vector shapes like rectangles, circles, and polygons using the shape tools or the Pen tool.
- Vector graphics are essential for creating icons, logos, and other scalable elements.



Color Palette:

- Figma lets you define and manage color palettes for your design project.
- You can save colors as swatches, making it easy to maintain consistency throughout your design.
- To use a color, you can apply it to shapes, text, or other design elements.



Practical: 09

AIM: Basic Clickable Prototyping using Figma. Creating a basic clickable prototype using Figma is a great way to demonstrate the user flow and interactions within your web or mobile app design. Follow Below Steps to create Click able Prototyping in Figma.

Step 1: Set Up Your Figma File

- Open Figma and create a new project or use an existing one.
- Create artboards or frames for each screen of your prototype. To do this, select the "Frame" tool from the toolbar on the left and draw rectangles to represent your screens.

Step 2: Design Your Screens

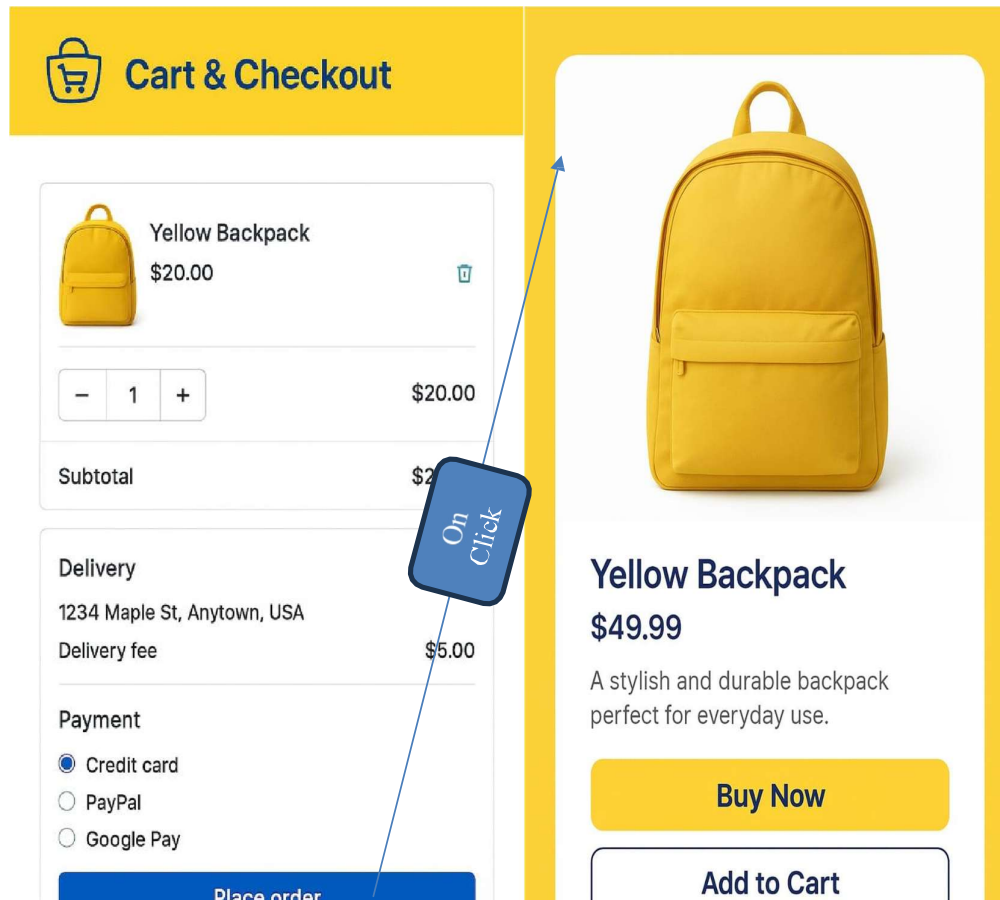
- Select the artboard/frame you want to design.
- Use Figma's design tools to create your user interface, including text, shapes, icons, and images. You can also import assets or use Figma's built-in design resources.
- Repeat this process for each screen in your prototype.

Step 3: Create Links Between Screens

- In Figma, you can create links between screens to simulate user interactions.
- Select an element (e.g., a button or text) on your first screen that you want to make clickable.
- In the right-hand panel, under the "Prototype" tab, click on the "Add Prototype" button.
- A blue arrow will appear next to your selected element. Click and drag the arrow to the target screen (the screen the user should navigate to when clicking the element).
- Configure the interaction settings:
 - i) Choose a trigger (e.g., "On Click").
 - ii) Set the destination screen.
 - iii) Define the animation type (e.g., "None" for a simple transition).
 - iv) You can also set delay, duration, and other options as needed.
- Repeat this process to create links between other screens and interactive elements.

Step 4: Preview Your Prototype

- To test your prototype, click the "Play" button in the top-right corner of Figma.
- Interact with your prototype by clicking on the elements you've linked. This will simulate the user experience and navigation.



Practical: 10

AIM: Create a Design System for an e-commerce app using Grid and Spacing, Typography, Color System, and UI elements like icons, images, buttons, Inputs, Cards, Search Bar, Lists, etc.

Design System for E-Commerce App:

1) Grid and Spacing:

i) Grid System:

Use a responsive grid system with 12 columns for layout.

Define breakpoints for different screen sizes (e.g., mobile, tablet, desktop).

Maintain consistent gutters and margins between elements for spacing.

ii) Spacing:

Define a spacing scale that includes values like 4px, 8px, 16px, 24px, and 32px.

Use spacing consistently for padding and margins between elements.

Maintain a baseline grid for text alignment.



2) Typography:

i) Headings:

Use a readable and modern typeface for headings (e.g., Roboto, Helvetica Neue).

Define various heading styles (H1, H2, H3) for different hierarchy levels.

Maintain consistent font sizes, line heights, and letter spacing.

ii) Body Text:

Choose a legible font for body text (e.g., Arial, Georgia).

Specify font size, line height, and paragraph spacing for readability. Use a limited number of text colors for standard and interactive text.

Font types

Font Family	Font Weight
ABeeZee	Regular
ABeeZee	Italic
Abyssinica SIL	Regular
SF Pro Text	Semibold

3) Color System:

i) Primary Colors:

Select a primary color palette that represents your brand (e.g., blue, orange). Define primary background and text colors for light and dark themes. Ensure color contrast meets accessibility standards.

ii) Secondary Colors:

Choose secondary colors to complement the primary palette. Use secondary colors for buttons, links, and accents. Maintain consistency in color usage across the app.

iii) Error and Success Colors:

Define error and success colors for feedback and validation messages.

Ensure these colors are distinguishable from the primary and secondary palettes.

 Button HEX: #db3022 HSL: 5, 73.1, 49.6 RGB: 219, 34, 48	 Text/Navigation HEX: #000000 HSL: 0, 0, 0 RGB: 0, 0, 0	 Other HEX: #f8e554 HSL: 37, 98.8, 66.3 RGB: 254, 84, 190
 Notification HEX: #2aa952 HSL: 139, 60.2, 41.4 RGB: 42, 82, 169	 Cards HEX: #999999 HSL: 0, 0, 60.8 RGB: 155, 155, 155	 Background HEX: #ffffff HSL: 0, 0, 100 RGB: 255, 255, 255

4) UI Elements:

i) Icons:

- Use a consistent icon set (e.g., Material Icons, Font Awesome). Define icon sizes and variations for different contexts.
- Ensure icons are legible and scalable.

ii) Images:

- Optimize images for fast loading times.
- Define image dimensions for various screen resolutions and sizes. Use lazy loading and responsive image techniques.

iii) Buttons:

- Create button styles for primary, secondary, and tertiary actions. Specify button sizes (small, medium, large) for different use cases. Include hover and active states for interactivity.

iv) Inputs:

- Design text input fields, text areas, and select boxes.
- Define input states (normal, hover, focus, error) with clear visual cues. Include placeholder text and labels for accessibility.

v) Cards:

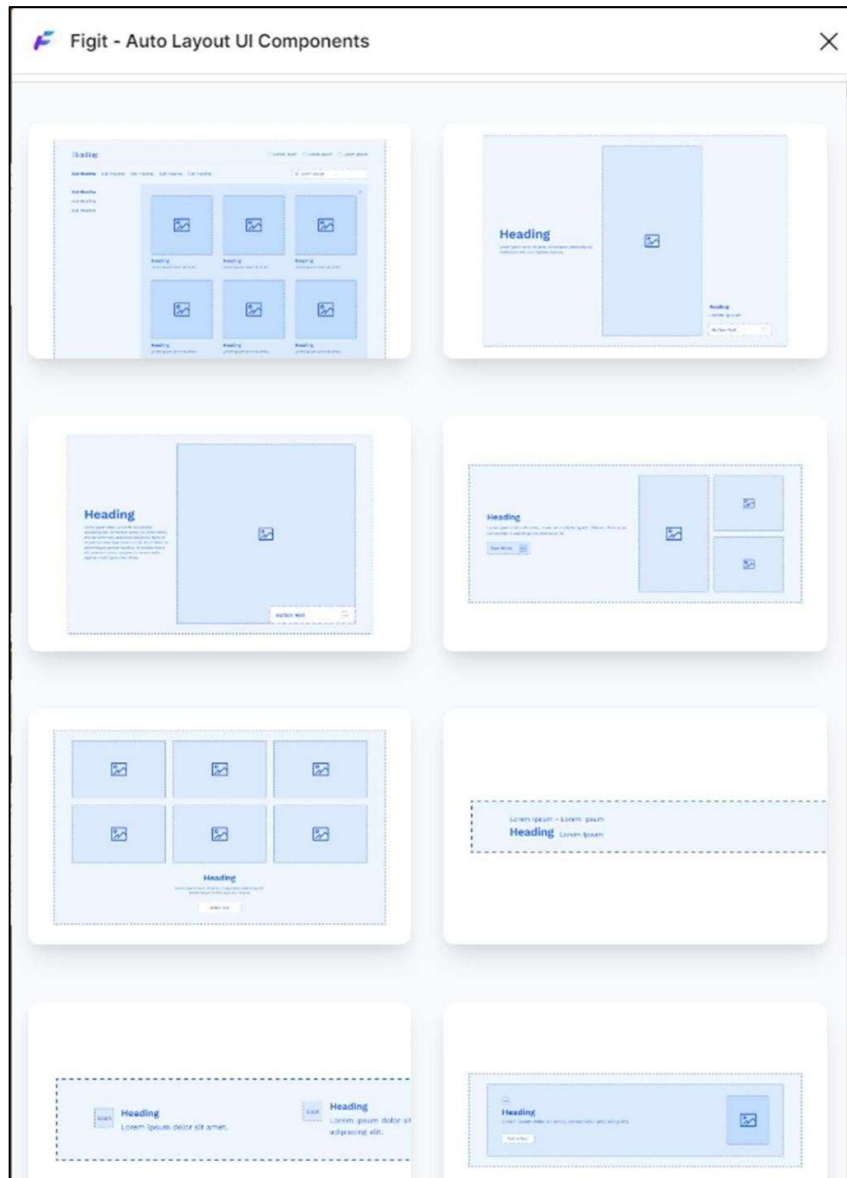
- Design product cards for listing items.
- Specify card variations for different product types (e.g., image, text, video). Include card actions (e.g., Add to Cart, Quick View).

vi) Search Bar:

- Create a search bar with clear input and search button styles. Include an autocomplete dropdown for search suggestions. Implement filters and sorting options where applicable.

vii) Lists:

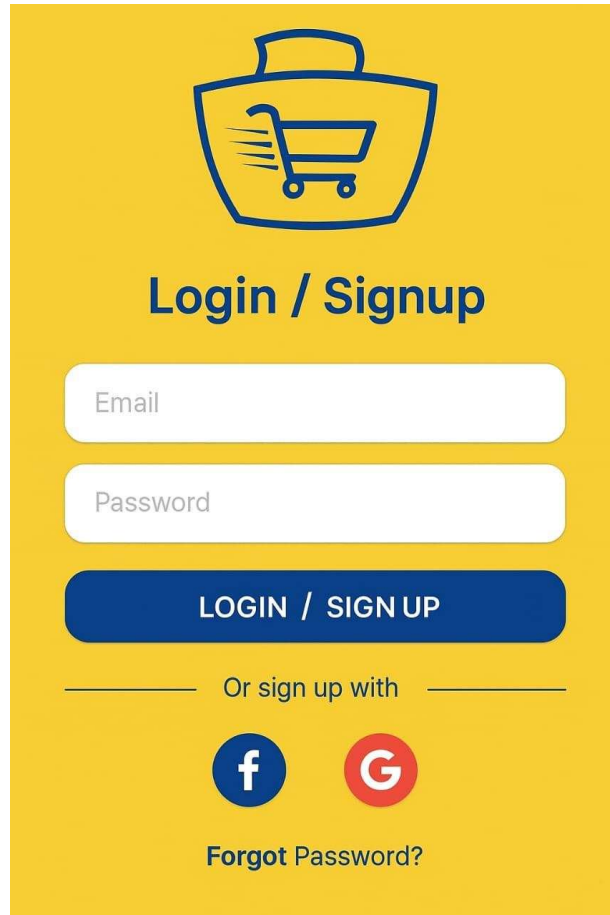
- Design lists for displaying products, categories, and reviews. Define list item styles, including spacing and typography.
- Include pagination or infinite scroll for long lists.



Practical-11

AIM: Reusing the above Design System, compose the Homepage, Product Page, and Checkout Page of the e-commerce app.

login Page:



The image shows a login/signup page with a yellow background. At the top is a blue icon of a shopping cart with a hand icon inside. Below the icon is the text "Login / Signup" in bold. There are two white input fields: "Email" and "Password". Below these is a blue button with the text "LOGIN / SIGN UP". Below the button is a horizontal line with the text "Or sign up with" in the center. Below this line are two circular icons: a blue Facebook "f" icon and a red Google "G" icon. At the bottom is the text "Forgot Password?" in bold.

order Page:



Yellow Backpack
\$49.99

A stylish and durable backpack
perfect for everyday use.

Buy Now

Add to Cart

Order tracking Page:

